

**AMENDMENT NO. 1 MARCH 2002
TO
IS 11833 : 1986 SPECIFICATION FOR DRY POWDER
FIRE EXTINGUISHER FOR METAL FIRES**

(*Page 3, clause 0.3*) — Insert the following new clauses after 0.3 and renumber the subsequent clause:

'0.4 A scheme for labelling environment friendly products known to be as ECO Mark has been introduced at the instance of the Ministry of Environment and Forests (MEF), Government of India. The ECO Mark would be administered by the Bureau of Indian Standards (BIS) under the *BIS Act*, 1986 as per the Resolution No. 71 dated 21 February 1991 and No. 425 dated 28 October 1992 published in the Gazette of the Government of India. For a product to be eligible for marking with ECO logo, it shall also carry the standard Mark of BIS besides meeting additional optional, environment friendly requirements. For this purpose, the Standard Mark of BIS would be a single mark being a combination of the ISI Mark and the ECO logo. Requirements to be satisfied for a product to qualify for the BIS Standard Mark for ECO friendliness being included in the relevant published standards through an amendment. These requirements are optional; manufacturing units are free to opt for the ISI Mark alone also.

The amendment is based on the Gazette Notification No. 160 dated 1 April 1999 for Fire Extinguishers as environment friendly products published in the Gazette of Government of India.'

(*Page 10, clause 12.2*) — Insert the following new clause after 12.2 and renumber the subsequent clauses:

'13 OPTIONAL REQUIREMENTS FOR ECO MARK

13.1 General Requirements

13.1.1 Any fire extinguisher having BIS Standard Mark qualifies for consideration of ECO Mark.

13.1.2 The products manufacturer must produce the consent clearance as per provision of the *Water (Prevention and Control of Pollution) Act*, 1974, *Water (Prevention and Control of Pollution) Cess Act*, 1977 and *Air (Prevention & Control of Pollution) Act*, 1981 respectively, along with authorization if required under *Environment (Protection) Act*, 1986, and the Rules made thereunder to the Bureau of Indian Standards while applying for ECO Mark.

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13.1.3 The products may display in brief the criteria based on which the product has been awarded ECO Mark.

13.1.4 The product may carry along with instructions for proper use so as to maximize product performance with statutory warning, if any, minimize waste and method of safe disposal.

13.1.5 The material used for product packaging (excluding refills) shall be recyclable, reusable or biodegradable.

13.1.6 The product must display a list of critical ingredients in descending order of quantity present in percent by weight. The list of such critical ingredients shall be identified by the Bureau of Indian Standards.

13.2 Specific Requirements

13.2.1 The fire extinguisher shall not contain any Ozone Depleting Substance (ODS) relevant to fire extinguishers industry as identified under the Montreal Protocol (Annex A).

13.2.2 Gas based extinguishing media once discharged in the atmosphere should not have atmospheric life time of more than a year (Annex B).

13.2.3 Chemical used should not have global warming potential (Annex C).

13.2.4 The metallic body and other metal parts of the fire extinguishers shall be free of lead or lead alloys.

13.2.5 The coating used for the metallic part shall not be formulated with mercury and mercury compounds or be tinted with pigments of lead, cadmium, chromium VI and their oxides. Excluded are natural impurities entailed by the production process up to the amount 0.1 percent by weight which are contained in the raw material.

NOTE — CO₂ extinguishers may be permitted till suitable substitutes are available.

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ANNEX A
(Clause 13.2.1)

**LIST OF OZONE DEPLETING SUBSTANCES (ODS) CONTROLLED
BY MONTREAL PROTOCOL**

<i>Trade Name</i>	<i>ODP</i>
Halon 1211	3.0
Halon 1301	10.0
Halon 2402	6.0
CFC-11	1.0
CFC-12	1.0
CFC-113	0.8
CFC-114	1.0
CFC-115	0.6
CCl ₄	1.1
C ₂ H ₃ C ₁₃	0.1
CFC-13	1.0
CFC-111	1.0
CFC-112	1.0
CFC-211	1.0
CFC-212	1.0
CFC-213	1.0
CFC-214	1.0
CFC-215	1.0
CFC-216	1.0
CFC-217	1.0
Methyl Bromide	0.6

NOTE — ODP values are relative to CFC-II which has been assigned arbitrary value of 1.0..

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ANNEX B
(clause 13.2.2)

LIST OF ATMOSPHERIC LIFE TIME OF GAS-BASED AGENTS

<i>Trade Name</i>	<i>Designation</i>	<i>Atmospheric Life Time (Year)</i>
Halon-13001	(CF 31)	<1 day
NAFS III	HCFC (Blend A)	12
FE 25	HCFC-125	36
FE 241	FCFC-124	6
FE 36	HFC-227 fa	250
FE 13	HFC-23	250
FM 200	HFC-227 EA	41
CEA 410	FC-3-1-10	2 600
Halon 1301	Halon 1301	65
Inergen	IG 541	—
Argonite	IG 55	—
Argon	IG 01	—

ANNEX C
(clause 13.2.3)

**LIST OF SUBSTANCES HAVING GLOBAL
WARMING POTENTIAL (GWP)**

<i>Trade Name</i>	<i>GWP (100 year) Vs CO₂</i>
Halon 1301	5 600
Inergen	—
Argonite	—
Argon	—
CEA 410	5 500
FM 200	3 300
FE 13	12 100
FE 36	8 000
FE 241	480
FE 25	3 200
NAFS III	1 450
CF 31	<5

(CED 22)